

Aliah University
Autumn Semester Examination - 2026
B.CA 3rd Year, 6th Semester Examination

Paper Name: Research Methodology
Paper Code: CSAUGPC35

Full Marks:80
Time: 3 hrs

Group A

1. Answer all questions (10 × 1 = 10)

I. Research is defined as:

- a) Random data collection b) Systematic investigation c) Guessing d) Observation only
[CO1 | PO1 | Remembering]

II. Which type of research uses numerical data?

- a) Qualitative b) Quantitative c) Mixed d) Exploratory
[CO1 | PO1 | Understanding]

III. Sampling frame refers to:

- a) Sample size b) Population list c) Data analysis d) Hypothesis
[CO2 | PO1 | Remembering]

IV. Census method involves:

- a) Partial study b) Complete enumeration c) Random sampling d) Hypothesis testing
[CO2 | PO1 | Understanding]

V. Null hypothesis represents:

- a) Significant relationship b) No relationship c) Strong effect d) Positive result
[CO3 | PO2 | Remembering]

VI. Which test is used for small samples?

- a) Z-test b) t-test c) Chi-square d) ANOVA
[CO3 | PO2 | Understanding]

VII. Sampling error occurs due to:

- a) Whole population study b) Sample selection c) Data analysis d) Report writing
[CO2 | PO2 | Understanding]

VIII. Qualitative research focuses on:

- a) Numbers b) Measurement c) Opinions and experiences d) Formulas
[CO1 | PO1 | Understanding]

IX. Level of significance is denoted by:

- a) β b) α c) μ d) σ
[CO3 | PO2 | Remembering]

X. ANOVA is used to:

- a) Compare two means b) Compare more than two means
c) Test variance only d) Collect data
[CO3 | PO2 | Understanding]

Group B
Answer any 4 question

4X5=20

2. Explain the characteristics of research and research methodology. [CO1 | PO1 | Understanding]
3. Differentiate between Quantitative and Qualitative research with examples. [CO1 | PO2 | Analyzing]
4. Explain sampling design and its importance in research. [CO2 | PO1 | Understanding]
5. Compare Census and Sample Survey with advantages and limitations. [CO2 | PO2 | Evaluating]
6. Explain different types of hypothesis with suitable examples. [CO3 | PO1 | Understanding]
7. Write a short note on sampling errors and non-sampling errors. [CO2 | PO2 | Understanding]

Group C
Answer any 5 question

5X10=50

8. a) Explain the steps in the research process with diagram. b) Discuss objectives and importance of research. [CO1 | PO2 | Applying + Understanding] (6+4=10)
9. a) Explain types of research based on purpose and approach. b) Compare quantitative, qualitative and mixed research methods. [CO1 | PO2 | Analyzing] (6+4=10)
10. a) Explain sampling techniques (probability and non-probability). b) Discuss factors affecting sample size. [CO2 | PO2 | Applying + Understanding] (6+4=10)
11. a) Explain hypothesis and its characteristics. b) Describe null and alternative hypothesis with examples. [CO3 | PO1 | Understanding] (6+4=10)
12. a) Explain steps of hypothesis testing with example. b) Discuss Type I and Type II errors. [CO3 | PO2 | Applying + Analyzing] (6+4=10)
13. a) Explain Z-test and t-test with suitable conditions. b) Describe Chi-square test and its applications. c) What is ANOVA? [CO3 | PO2 | Analyzing + Understanding] (4+3+3=10)
14. Write short notes on any two:
a) Motivation in research
b) Variables in research
c) Sampling frame
d) Limitations of hypothesis testing
[CO1/2/3 | PO1 | Understanding] (2×5=10)